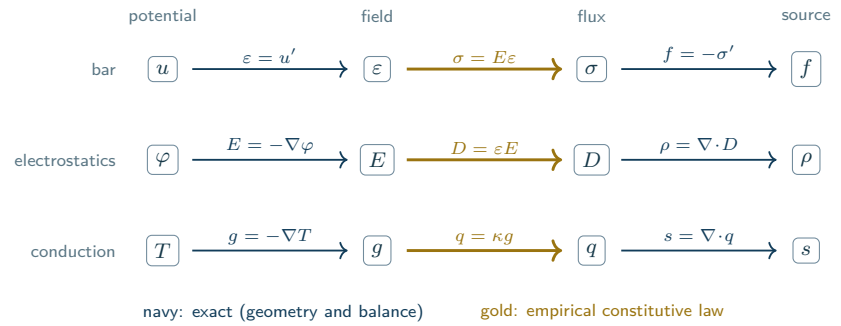
PHYSICS ON ONE PAGE

Field equations and constitutive laws

Balance laws say what is conserved, and empirical constitutive laws supply the material or force that closes them.

1 / THE RULE AND THE PICTURE

Elasticity, electrostatics and heat conduction share one chain with one empirical link



In each row a potential gives a field by geometry, a source follows from a flux by balance, and the middle arrow, from field to flux, is the one that depends on the material. In the bar three unknown fields u, ε, σ face two equations until $\sigma = E\varepsilon$ is added:

balance + geometry + constitutive law = closed

2 / CONSTITUTIVE LAWS AS EMPIRICAL CLOSURES

Each law is measured, holds in a range, and is added to a balance equation that cannot supply it

empirical law	statement	closes	stops working
Hooke	$\sigma = E\varepsilon$, $F = -kx$	equilibrium of a bar	beyond the elastic limit; mild steel yields near 0.12% strain
ideal gas	$pV = nRT$	fluid balance equations	CO_2, 1 mol in 1 L at 300 K: 24.9 bar ideal, 22.4 bar van der Waals
gravity	$F = \frac{GMm}{r^2}$	$F = ma$ for planets	in strong fields: Mercury's perihelion advances 43 arcseconds per century beyond it
Ohm	$J = \sigma E$	charge balance	in diodes and other nonlinear conductors
Fourier	$q = -\kappa\nabla T$	heat balance	over distances near the mean free path of the carriers
dielectric	$D = \varepsilon E$	Gauss's law	near breakdown, about 3 MV/m in air

3 / NEWTON'S SECOND LAW AS A TEMPLATE

$F = \frac{dp}{dt}$ predicts nothing until a force law is supplied

The second law says that momentum changes at the rate of the force and leaves F unspecified, so each force law below gives a different motion from the same statement. What carries over from case to case is the balance of momentum, whose conservation follows from translation symmetry, and what changes is the empirical force law.

force law supplied	equation of motion	solution	example
$F = -kx$	$m\ddot{x} = -kx$	$x = A\cos(\omega t + \phi)$, $\omega = \sqrt{k/m}$	$k = 200$ N/m, $m = 2$ kg: $\omega = 10$ rad/s
$F = -\frac{GMm}{r^2}$	$m\ddot{\mathbf{r}} = -\frac{GMm}{r^2}\hat{\mathbf{r}}$	$T^2 = \frac{4\pi^2 a^3}{GM}$	Earth, $a = 1.496 \times 10^{11}$ m: 365.3 days
$F = -bv$	$m\dot{v} = -bv$	$v = v_0 e^{-bt/m}$	$m = 2$ kg, $b = 0.5$ kg/s: decay time 4 s
$F = -mg$	$m\ddot{x} = -mg$	$x = \frac{1}{2}gt^2$ falling from rest	one second of fall covers 4.9 m

4 / MAXWELL'S EQUATIONS NEED CONSTITUTIVE RELATIONS

Two curl equations and six material relations determine twelve field components

$$\nabla\cdot\mathbf{D} = \rho, \qquad \nabla\cdot\mathbf{B} = 0, \qquad \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}, \qquad \nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t}$$

with the constitutive relations $\mathbf{D} = \varepsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$ and $\mathbf{J} = \sigma\mathbf{E}$. The two curl equations are 6 scalar equations for the 12 components of $\mathbf{E}, \mathbf{B}, \mathbf{D}, \mathbf{H}$, and the six relations $\mathbf{D} = \varepsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$ supply the rest, while the divergence equations are constraints that persist in time when $\nabla\cdot\mathbf{J} + \frac{\partial\rho}{\partial t} = 0$. The relations are measured and depend on frequency: water has $\varepsilon_r \approx 80$ for static fields and $n^2 = 1.78$ at optical frequencies.

5 / WHERE THE CONSTITUTIVE LAW SHOWS UP

Every wave speed is a stiffness from a constitutive law over an inertia from a balance law

wave in	speed	inputs	value
steel bar	$\sqrt{E/\rho}$	$E = 200$ GPa, $\rho = 7850$ kg/m^3	5.05 km/s
air	$\sqrt{\gamma p/\rho} = \sqrt{\gamma RT}$	$\gamma = 1.4$, $T = 293$ K	343 m/s
stretched string	$\sqrt{T/\mu}$	$T = 100$ N, $\mu = 0.01$ kg/m	100 m/s
vacuum	$1/\sqrt{\varepsilon_0\mu_0}$	measured ε_0 and μ_0	2.998×10^8 m/s

Euler's equations for a gas, balance of mass, momentum and energy, are 5 equations for the 6 unknowns ρ, $\mathbf{v}$, p, e, and the equation of state, for air $p = (\gamma - 1)\rho e$, closes them and only then gives a sound speed.

ASIDE / TRY THESE

Use $a = 3.64$ L^{2}bar/mol^2 and $b = 0.0427$ L/mol in $p = \frac{RT}{V-b} - \frac{a}{V^2}$ to check 22.4 bar. Count the unknowns and equations for a heat-conducting bar.

LINKS TO OTHER SHEETS

The work and impulse sheet treats $F = \frac{dp}{dt}$ as an integral over a window, and the Fourier sheet treats the wave equations behind these speeds.

Convention. ε strain in the bar and permittivity in the fields, by context; E Young's modulus or electric field, by context; all numbers were checked by computation.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames